

First Order Difference Equation:

A difference equation expresses a relationship between a dependent variable and a lagged independent variable (or variables) which changes at discrete intervals of time, for example, $I_t = f(y_{t-1})$ where I and Y are measured at the end of each year. The order of a difference equation is determined by the greatest number of periods lagged. A first order difference equation expresses a time lag of one period; a second order, two periods; etc. the change in y as t changes from t to t+1 is called the first difference of y. it is written

$$\frac{\Delta y}{\Delta t} = \Delta y_t = y_{t+1} - y_t$$

Where Δ is an operator

replacing $\frac{d}{dt}$ that is used to measure the continuous change in differential equations. The solution of difference equation defines y for every value of t and does not contain a difference equation.

Solving a First Order Difference Equation:

There are two methods (given in book) for solving a first difference equation

- (i) Iterative method
- (ii) General method

Iterative method:

Given that the initial value of y is $y_{(0)}$ in the difference equation

$$y_{(t)} = by_t \rightarrow (a)$$

A solution is found as follows. By successive substitutions of $t=0, 1, 2, 3, \text{ etc}$ in (a)

$$y_{(t)} = by_t$$

$$y_1 = by_0$$

$$y_2 = by_1 = b(by_0) = b^2 y_0$$

$$y_3 = by_2 = b(b^2 y_0) = b^3 y_0$$

$$y_4 = by_3 = b(b^3 y_0) = b^4 y_0$$

Thus for any period t

$$y_t = b^t y_0$$

This method is called the iterative method. Since y_0 is a constant notice the crucial role b plays in determining values for y as t changes.

Example 1:

Find the solution of the difference equation

$$\Delta y_t = 2 \quad y_0 = 15$$

$$\text{As } \Delta y_t = y_{t+1} - y_t = 2$$

$$y_{t+1} = 2 + y_t$$

Substituting $t=0, 1, 2, 3, \text{ etc}$

$$y_1 = y_0 + 2$$

$$y_2 = y_1 + 2 = (y_0 + 2) + 2 = y_0 + 2(2)$$

$$y_3 = y_2 + 2 = [y_0 + 2(2)] + 2 = y_0 + 3(2)$$

And in general, for any period t

$$y_t = y_0 + t(2) = 15 + 2t$$

Example 2:

Solve the difference equation

$$y_{t+1} - 9y_t = 0$$

$$y_{t+1} = 9y_t$$

Substituting successively $t=0, 1, 2, 3, \text{ etc}$

$$y_1 = 0.9y_0$$

$$y_2 = 0.9y_1 = 0.9(0.9y_0) = 0.9^2 y_0$$

$$y_3 = 0.9y_2 = 0.9(0.9^2 y_0) = (0.9)^3 y_0$$

...

These can be summarized into the solution

$$y_t = (0.9)^t y_0$$

Example 3:

Solve the homogenous difference equation

$$my_{t+1} - ny_t = 0$$

$$y_{t+1} = \frac{n}{m} y_t$$

Substituting successively $t=0, 1, 2, 3, \text{ etc}$

$$y_1 = \frac{n}{m} y_0$$

$$y_2 = \frac{m}{m} y_1 = \frac{n}{m} \left(\frac{n}{m} y_0 \right) = \left(\frac{n}{m} \right)^2 y_0$$

$$y_3 = \frac{n}{m} y_2 = \frac{n}{m} \left(\left(\frac{n}{m} \right)^2 y_0 \right) = \left(\frac{n}{m} \right)^3 y_0$$

$$y_4 = \frac{n}{m} y_3 = \frac{n}{m} \left(\left(\frac{n}{m} \right)^3 y_0 \right) = \left(\frac{n}{m} \right)^4 y_0$$

$$\vdots$$

$$y_t = \left(\frac{n}{m} \right)^t y_0$$

General method:

Suppose that we are seeking the solution to the first order difference equation

$$y_{t+1} + ay_t = c$$

Where a and c are two constants. The general solution will consist of the two components: a particular integral y_p and a complementary function y_c

Complementary Function:

Complementary function represents the deviation of the time path from that equilibrium (inter temporal). As we know that y_c is the general solution of the reduced equation, i.e.

$$y_{t+1} + ay_t = 0$$

Try a simplest possible type of solution of the form

$$y_t = Ab^t \quad (\text{where } Ab^t \neq 0)$$

$$\text{Then } y_{t+1} = Ab^{t+1}$$

Substituting into above reduced equation

$$Ab^{t+1} + aAb^t = 0$$

$$Ab^t(b+a) = 0$$

Dividing both sides by Ab^t

$$b+a=0$$

$$b=-a$$

$$\text{Hence } y_c (= Ab^t) = A(-a)^t$$

Particular Integral:

The y_p component represents the inter temporal equilibrium level of y . It is the any solution of the complete equation, i.e.

$$y_{t+1} + ay_t = c$$

Try a solution of the simplest form

$$y_t = k \quad (k = \text{constant})$$

It follows

$$y_{t+1} = k$$

Substituting into above

$$k + ak = c$$

$$k(1+a) = c$$

$$y_p (= k) = \frac{c}{1+a} \quad (a \neq -1)$$

Since it is a constant it represents a stationary equilibrium. If it happens that $a = -1$, then try a solution of the form

$$y_t = kt$$

$$\text{Then } y_{t+1} = k(t+1)$$

Substituting into above

$$k(t+1) + akt = c$$

$$kt + k + akt = c$$

$$k(t+1+at) = c$$

$$k = \frac{c}{t+1+at} = c \quad (\text{because } a = -1)$$

$$\text{Hence } y_p (= kt) = ct$$

Since this particular integral is non constant function of t , it represents a moving equilibrium.

General Solution:

By adding y_c and y_p together, we may write the general solution in one of the two following forms.

$$y_t = A(-a)^t + \frac{c}{1+a} \rightarrow (i) \quad (a \neq -1)$$

$$y_t = A(-a)^t + ct = A + ct \rightarrow (ii) \quad (a = -1)$$

Definite solution:

In order to definitize the solution set $t = 0$

$$y_0 = A + \frac{c}{1+a}$$

$$A = y_0 - \frac{c}{1+a}$$

Substituting into (i)

$$y_t = \left[y_0 - \frac{c}{1+a} \right] (-a)^t + \frac{c}{1+a} \quad \text{case of } (a \neq -1)$$

For (ii) set $t = 0$

$$y_{(0)} = A + c(0) = A$$

$$A = y_{(0)}$$

Hence

$$y_t = y_0 + ct \quad (\text{definite solution case of } a = -1)$$

Example:

Solve the first order difference equation

$$y_{t+1} - 5y_t = 1 \quad y_0 = \frac{7}{4}$$

$$\text{Let } y_t = Ab^t$$

$$\text{Then } y_{t+1} = Ab^{t+1}$$

$$\text{Hence } Ab^t - 5Ab^t = 0$$

$$Ab^t(b-5) = 0$$

$$b = 5$$

$$y_c (= Ab^t) = A(5)^t$$

For y_p

$$\text{Let } y_t = k \text{ then } y_{t+1} = k$$

Substituting into given equation

$$k - 5k = 1$$

$$k(1-5) = 1$$

$$k = \frac{1}{1-5} = \frac{1}{-4}$$

$$y_p = \frac{1}{-4} \quad (\text{when } a \neq -1)$$

Adding y_c and y_p

$$y_t = y_c + y_p$$

$$y_t = A(5)^t + \left(-\frac{1}{4} \right) = A(5)^t - \frac{1}{4}$$

General Solution

Set $t = 0$ in order to definitize the solution

$$y_0 = A(5)^0 - \frac{1}{4}$$

$$y_0 + \frac{1}{4} = A$$

$$y_t = \left(y_0 + \frac{1}{4} \right) (5)^t - \frac{1}{4}$$

$$y_t = \left(\frac{7}{4} + \frac{1}{4} \right) (5)^t - \frac{1}{4}$$

$$y_t = 2(5)^t - \frac{1}{4} \quad \text{Definite solution}$$

Exercise 16.2:

Q.1. Convert the following difference equations into the form of (16.2')

$$\{y_{t+1} = y_t + 2 \rightarrow (16.2'')\}$$

$$(a) \Delta y_t = 7$$

$$y_{t+1} - y_t = 7 \quad \Delta y_t = y_{t+1} - y_t$$

$$y_{t+1} = y_t + 7$$

$$(b) \Delta y_t = 0.3 y_t$$

$$y_{t+1} - y_t = 0.3 y_t$$

$$y_{t+1} = 0.3 y_t + y_t$$

$$y_{t+1} = 1.3 y_t$$

$$(c) \Delta y_t = 2 y_t - 9$$

$$y_{t+1} - y_t = 2 y_t - 9$$

$$y_{t+1} = 2 y_t + y_t - 9$$

$$y_{t+1} = 3 y_t - 9$$

Q.2. Solve the following difference equation by iteration

$$(a) y_{t+1} = y_t + 1 \quad y_0 = 10$$

Substituting $t = 0, 1, 2, 3, \text{ etc}$

$$y_1 = y_0 + 1$$

$$y_2 = y_1 + 1 = y_0 + 1 + 1 = y_0 + 2(1)$$

$$y_3 = y_2 + 1 = y_0 + 2 + 1 = y_0 + 3(1)$$

$$y_4 = y_3 + 1 = y_0 + 3 + 1 = y_0 + 4(1)$$

$$\vdots$$

$$y_t = y_0 + t(1)$$

$$y_t = 10 + t$$

$$(b) y_{t+1} = \alpha y_t \quad y_0 = \beta$$

Substituting $t = 0, 1, 2, 3, \text{ etc}$

$$y_1 = \alpha y_0$$

$$y_2 = \alpha y_1 = \alpha(\alpha y_0) = \alpha^2 y_0$$

$$y_3 = \alpha y_2 = \alpha(\alpha^2 y_0) = \alpha^3 y_0$$

$$y_4 = \alpha y_3 = \alpha(\alpha^3 y_0) = \alpha^4 y_0$$

$$\vdots$$

$$y_t = \alpha^t y_0 \quad \text{As } y_0 = \beta$$

$$y_t = \beta \alpha^t$$

Or

$$y_t = \alpha^t \beta$$

$$(c) y_{t+1} = \alpha y_t - \beta \quad y_t = y_0 \text{ when } t=0$$

Substituting successively $t=0, 1, 2, 3, \text{ etc}$

$$y_1 = \alpha y_0 - \beta$$

$$y_2 = \alpha y_1 - \beta = \alpha(\alpha y_0 - \beta) - \beta = \alpha^2 y_0 - \alpha\beta - \beta$$

$$y_3 = \alpha y_2 - \beta = \alpha(\alpha^2 y_0 - \alpha\beta - \beta) - \beta = \alpha^3 y_0 - \alpha^2\beta - \alpha\beta - \beta$$

$$y_4 = \alpha y_3 - \beta = \alpha(\alpha^3 y_0 - \alpha^2\beta - \alpha\beta - \beta) - \beta$$

$$\vdots$$

$$= \alpha^4 y_0 - \alpha^3\beta - \alpha^2\beta - \alpha\beta - \beta$$

$$y_t = \alpha^t y_0 - \beta(1 + \alpha + \alpha^2 + \alpha^3 + \dots + \alpha^{t-1})$$

Applying formula for finite series when $r < 1$

$$y_t = \alpha^t y_0 - \beta \left[\frac{a(1-r^n)}{1-r} \right] = \alpha^t y_0 - \beta \left[\frac{1(1-\alpha^t)}{1-\alpha} \right]$$

$$y_t = \alpha^t y_0 - \beta \frac{(1-\alpha^t)}{1-\alpha}$$

Q.3. Rewrite the equation in preceding problem in the form of (16.6) and solve by applying formula (16.8') or 16.9' whichever is appropriate. Does your answer check with those obtained by iterative method?

$$(a) y_{t+1} = y_t + 1$$

$$y_{t+1} - y_t = 1 \quad a = -1$$

$$c = 1$$

Applying formula

Since $a = -1$ we can apply the following formula

$$y_t = y_0 + ct$$

$$y_t = y_0 + t$$

$$y_t = 10 + t \quad \text{As } (y_0 = 10)$$

This is checked with that obtained by iterative method

$$(c) y_{t+1} = \alpha y_t - \beta$$

$$y_{t+1} - \alpha y_t = -\beta$$

$$a = -\alpha$$

$$c = -\beta$$

$$y_t = \left[y_0 - \frac{c}{1+a} \right] (-a)^t + \frac{c}{1+a}$$

$$y_t = \left[y_0 + \frac{\beta}{1-\alpha} \right] (\alpha)^t - \frac{\beta}{1-\alpha} = y_0 \alpha^t + \frac{\beta}{1-\alpha} (\alpha)^t - \frac{\beta}{1-\alpha}$$

$$y_t = y_0 \alpha^t - \frac{\beta(1-\alpha^t)}{1-\alpha} = y_0 \alpha^t - \frac{\beta(1-\alpha^t)}{1-\alpha}$$

Q.4. For each of the following difference equation, use the procedure illustrated in the derivation of (16.8') and (16.9') to find y_c , y_p and the definite solution

$$(a) y_{t+1} + 3y_t = 4 \quad y_0 = 4$$

Finding y_c

As we know that complementary function is the general solution of the reduced equation, i.e.

$$y_{t+1} + 3y_t = 0$$

Try a solution of the form

$$y_t = Ab^t \quad (Ab^t \neq 0)$$

Which implies that

$$y_{t+1} = Ab^{t+1}$$

Substituting into above homogenous or reduced equation

$$y_{t+1} + 3y_t = 0$$

$$Ab^{t+1} + 3Ab^t = 0$$

$$Ab^t(b+3) = 0$$

$$b = -3$$

$$y_c = Ab^t = A(-3)^t$$

For y_p

$$\text{Let } y_t = k \text{ which follows } y_{t+1} = k$$

Substituting into given non homogenous or complete equation, i.e.

$$y_{t+1} + 3y_t = 4$$

$$k + 3k = 4$$

$$k(1+3) = 4$$

$$y_p = k = \frac{4}{4} = 1$$

By adding y_c and y_p we get general solution

$$y_t = y_c + y_p$$

$$y_t = A(-3)^t + 1 \rightarrow \text{General Solution}$$

Set $t = 0$ in order to definitize the solution

$$y_0 = A(-3)^0 + 1$$

$$y_0 - 1 = A$$

$$4 - 1 = A$$

$$A = 3$$

$$y_t = 3(-3)^t + 1 \quad \text{Definite solution}$$

$$(b) 2y_{t+1} - y_t = 6 \quad y_0 = 7$$

Dividing both sides by 2

$$y_{t+1} - \frac{1}{2}y_t = 3$$

To find y_c

$$y_{t+1} - \frac{1}{2}y_t = 0$$

To find y_c trying a solution $y_t = Ab^t$ which implies that $y_{t+1} = Ab^{t+1}$

$$Ab^{t+1} - \frac{1}{2}Ab^t = 0$$

$$Ab^t \left(b - \frac{1}{2} \right) = 0$$

$$b = \frac{1}{2}$$

$$\text{Hence } y_c = A \left(\frac{1}{2} \right)^t$$

To find y_p let $y_t = k$

This implies that $y_{t+1} = k$

Substituting into above non homogenous equation, we get

$$k - \frac{1}{2}k = 3$$

$$k \left(1 - \frac{1}{2} \right) = 3$$

$$\frac{1}{2}k = 3$$

$$k = 3 \times 2 = 6$$

$$\text{Hence } y_p = k = 6$$

$$y_p = 6$$

$$y_t = y_c + y_p$$

$$y_t = A \left(\frac{1}{2} \right)^t + 6$$

Set $t = 0$

$$y_0 = A \left(\frac{1}{2} \right)^0 + 6$$

$$7 - 6 = A$$

$$A = 1$$

$$y_t = \left(\frac{1}{2} \right)^t + 6 \quad \text{Definite solution}$$

$$(c) y_{t+1} = 0.2y_t + 4 \quad (y_0 = 4)$$

$$y_{t+1} - 0.2y_t = 4$$

To find y_c

$$y_{t+1} - 0.2y_t = 0$$

$$\text{Let } y_t = Ab^t$$

$$y_{t+1} = Ab^{t+1}$$

$$Ab^{t+1} - 0.2Ab^t = 0$$

$$Ab^t(b - 0.2) = 0$$

$$b = 0.2$$

$$\text{Hence } y_c = A(0.2)^t$$

Finding y_p

$$\text{Let } y_t = k$$

$$y_{t+1} = k$$

Substituting into above non homogenous equation

$$k - 0.2k = 4$$

$$k(1 - 0.2) = 4$$

$$k = \frac{4}{1 - 0.2} = \frac{4}{0.8} = 5$$

$$\text{Hence } y_p = 5$$

By adding y_c and y_p we get general solution

$$y_t = y_c + y_p$$

$$y_t = A(0.2)^t + 5 \quad \text{General solution}$$

To definitize the solution set $t = 0$

$$y_0 = A(0.2)^0 + 5$$

$$4 - 5 = A$$

$$A = -1$$

$$y_t = -(0.2)^t + 5 \quad \text{Definite solution}$$

The Dynamic Stability of Equilibrium:

In period analysis, the dynamic stability of equilibrium depends upon the Ab^t term in the complementary function.

The Significance of b:

Whether the equilibrium is dynamically stable is a question of whether or not the complementary function will tend to zero as $t \rightarrow \infty$. Basically we must analyze the path of the term Ab^t as t is increased indefinitely. Obviously the value of b is of the crucial importance in this regard. Let us first consider its significance alone, by disregarding the co-efficient A (by assuming $A = 1$)

For analytical purpose, we can divide the range of possible values of $b, (-\infty, +\infty)$, into seven distinct region as shown in the following table.

A Classification of the Valued of b:

Value of b^t in different time periods:

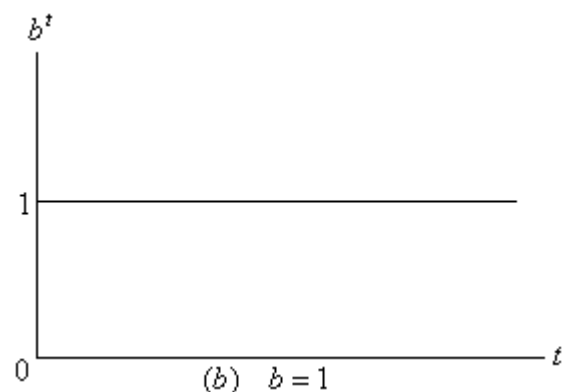
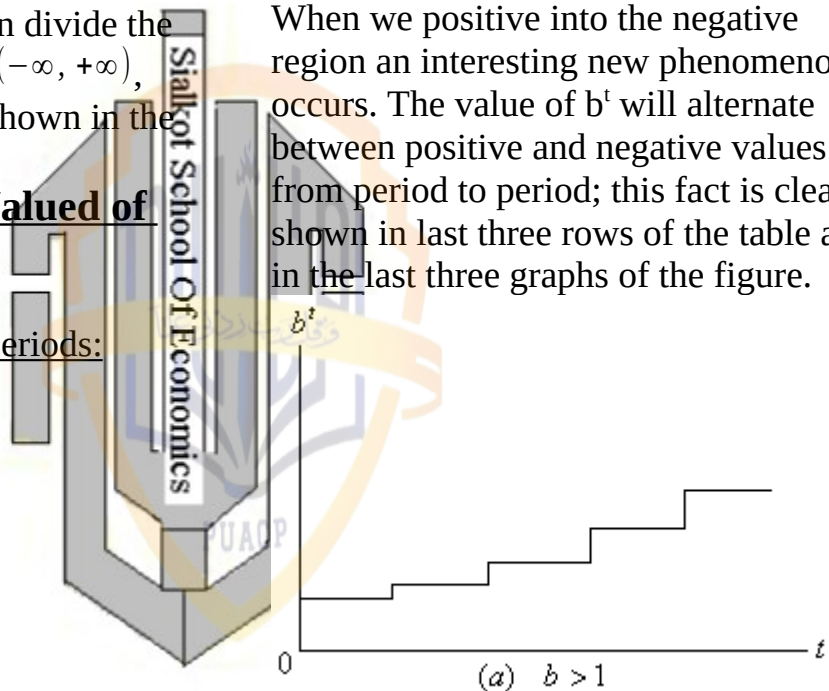
In each region the exponential expression b^t generates a different type of time path. These are exemplified in the above table and illustrated in the figure. In region 1 (where $b > 1$) b^t must increase with t at an increasing pace, thus moving farther and farther away from the horizontal axis. Its graph is shown as a step function rather than as a smooth curve; this is because we are dealing with a period analysis.

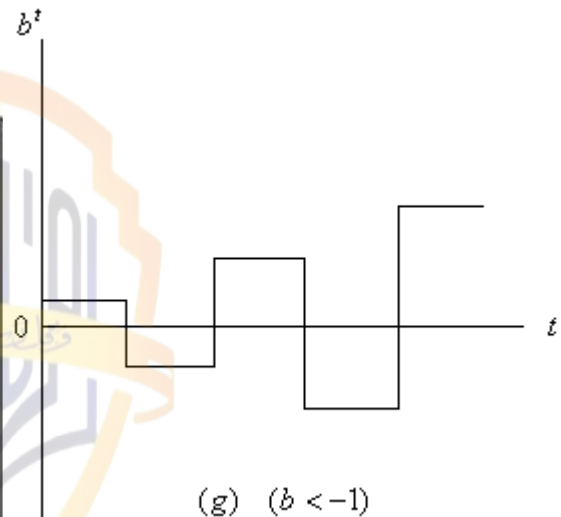
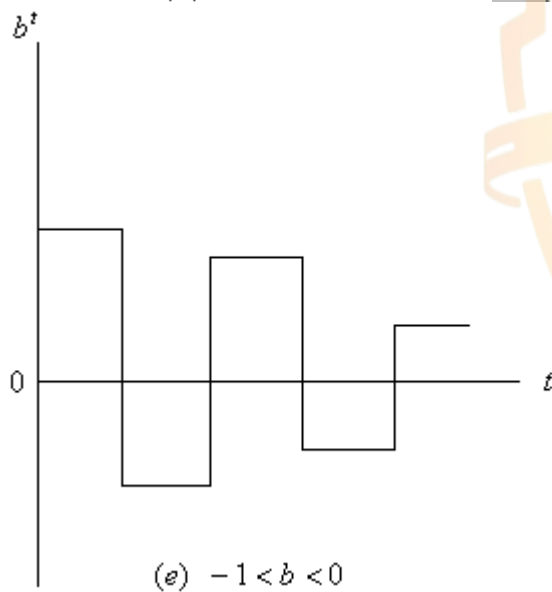
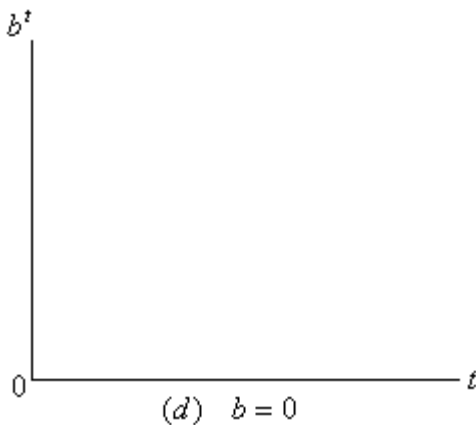
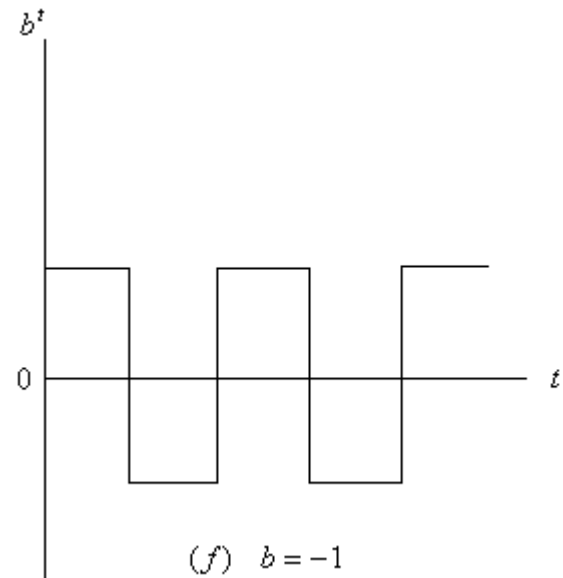
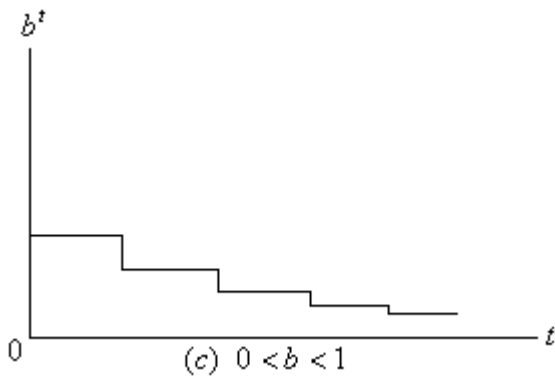
In region II ($b = 1$), b^t will remain at unity for all the values of t . its graph will

thus be a horizontal straight line. In region III ($0 < b < 1$), b^t represents a positive fraction raised to integer powers. As the power is increased b^t must decrease (though it will always remain positive) thus drawing closer and closer to the horizontal axis.

In region IV ($b = 0$), $b^t = 0$ for all the values of t . so its graph will coincide with the horizontal axis. However this case is of the peripheral (interest only since we have earlier adopted the assumption that $Ab^t \neq 0$ which implies $b \neq 0$)

When we positive into the negative region an interesting new phenomenon occurs. The value of b^t will alternate between positive and negative values from period to period; this fact is clearly shown in last three rows of the table and in the last three graphs of the figure.





In region V ($-1 < b < 0$) where b is a negative fraction, the alternating time path tends to get closer and closer to the horizontal axis.

In region VI ($b = -1$), a perpetual alternation between $+1$ and -1 results.

In region VII ($b < -1$), the alternating time path will deviate farther and farther from the horizontal axis.

The essence of the above discussion can be conveyed in the following general statement; the time path of b^t ($b \neq 0$) will be

Nonoscillatory if $|b| < 1$

Divergent if $|b| > 1$

The Rule of A:

There are two effects of A which are relatively easy to take into account. First the magnitude of A can serve to “blow up” (if, say, $A=3$) or “prove down” (if, say, $A=1/5$) the values of b^t . that is, it can produce a scale effect without changing the basic configuration of the time path. The sign of A on the other hand does materially affect the shape of the path because, if b^t is multiplied by $A=-1$ then each time path show in the above figure, will be replaced by its own mirror image with reference to the horizontal axis. Thus a negative A can produce a mirror effect as well as scale effect.

Note:

When $b=1$, the time path is to be considered divergent not because of the appearance of t in the particular integral (as $b=-a$) but because with a non zero A, there will be a constant deviation from the moving equilibrium (i.e. $y_p = ct$)

Example 1:

What kind of time path is represented by

$$y_t = 2\left(-\frac{4}{5}\right)^t + 9 \quad ? \text{ Since } b = -\frac{4}{5} < 0$$

the time path is oscillatory. But since $|b| = \frac{4}{5} < 1$ the oscillation is damped and the time path converges to the equilibrium level of 9. You should exercise care not to confuse

$2\left(-\frac{4}{5}\right)^t$ with $-2\left(\frac{4}{5}\right)^t$; they represent entirely different time path configuration.

Example 2:

How do you characterize time path

$$y_t = 3(2)^t + 4 ?$$

Since $b=2>1$, the time path will diverge from the equilibrium level of 4.

Exercise 16.3:

Q.1. Discuss the nature of the following time paths:

(a) $y_t = 3^t + 1$

Since $b=3>0$ no oscillation will occur

And $|b|=3>1$ the time path will diverge from equilibrium level of 1

(b) $y_t = 3\left(\frac{1}{3}\right)^t$

Since $b = \frac{1}{3} > 0$

no oscillation will occur

And $|b| = \frac{1}{3} < 1$ the time path will converge

(c) $y_t = 5\left(-\frac{1}{10}\right)^t + 3$

Since $b = -\frac{1}{10} < 0$

the time is oscillatory

$|b| = \frac{1}{10} < 1$

The time path will converge to the equilibrium level of 3.

(d) $y_t = -3\left(\frac{1}{4}\right)^t + 2$

Since $b = \frac{1}{4} > 0$

no oscillation will occur

$|b| = \frac{1}{4} < 1$

The time path will converge to the equilibrium level of 2.

Q.2. what is the nature of the time path obtained from each of the difference equation in Exercise 16.2–4

(a) $y_t = 3(-3)^t + 1$

Since $b=-3<0$ the time path is oscillatory

$|b|=3>1$ the time path will diverge from equilibrium level of 1.

(b) $y_t = \left(\frac{1}{2}\right)^t + 6$

Since $b = \frac{1}{2} > 0$ no oscillation will occur
 $|b| = \frac{1}{2} < 1$ The time path will converge to the equilibrium level of 6.

$$(c) y_t = -(0.2)^t + 5$$

Since $b = 0.2 > 0$ no oscillation will occur
 $|b| = 0.2 < 1$ time path will converge to the level of 5.

Q.3. Find the solution of the following and determine whether the time path is oscillatory and convergent.

$$(a) y_{t+1} - \frac{1}{3} y_t = 6 \quad (y_0 = 1)$$

$$a = -\frac{1}{3} \quad c = 6$$

Applying formula

$$y_t = \left[y_0 - \frac{c}{1+a} \right] (-a)^t + \frac{c}{1+a}$$

$$y_t = \left[1 - \frac{6}{1 - \frac{1}{3}} \right] \left(\frac{1}{3} \right)^t + \frac{6}{1 - \frac{1}{3}}$$

$$y_t = \left[1 - \frac{6}{\frac{2}{3}} \right] \left(\frac{1}{3} \right)^t + \frac{6}{\frac{2}{3}}$$

$$y_t = \left[1 - \frac{18}{2} \right] \left(\frac{1}{3} \right)^t + \frac{18}{2}$$

$$y_t = [1 - 9] \left(\frac{1}{3} \right)^t + 9$$

$$y_t = -8 \left(\frac{1}{3} \right)^t + 9$$

Since $b = \frac{1}{3} > 0$ no oscillation will occur

$|b| = \frac{1}{3} < 1$ Time path will converge to the equation level of 9.

$$(b) y_{t+1} + 2 y_t = 9 \quad y_0 = 4$$

$$a = 2 \quad c = 9$$

$$y_t = \left[y_0 - \frac{c}{1+a} \right] (-a)^t + \frac{c}{1+a}$$

$$y_t = \left[4 - \frac{9}{1+2} \right] (-2)^t + \frac{9}{1+2}$$

$$y_t = [4 - 3] (-2)^t + 3$$

$$y_t = (-2)^t + 3$$

$b = -2 < 0$ Oscillation will occur

$|b| = 2 > 1$ Time path will divergent from the equation level of 3.

$$(c) y_{t+1} + \frac{1}{4} y_t = 5 \quad (y_0 = 2)$$

$$a = \frac{1}{4}; \quad c = 5$$

$$y_t = \left[y_0 - \frac{c}{1+a} \right] (-a)^t + \frac{c}{1+a}$$

$$y_t = \left[2 - \frac{5}{1 + \frac{1}{4}} \right] \left(-\frac{1}{4} \right)^t + \frac{5}{1 + \frac{1}{4}}$$

$$y_t = \left[2 - \frac{5}{\frac{5}{4}} \right] \left(-\frac{1}{4} \right)^t + \frac{5}{\frac{5}{4}}$$

$$y_t = [2 - 4] \left(-\frac{1}{4} \right)^t + 4$$

$$y_t = -2 \left(-\frac{1}{4} \right)^t + 4$$

$b = -\frac{1}{4} < 0$ Oscillation will occur

$|b| = \frac{1}{4} < 1$ Time path will converge to the equilibrium level of 4.

$$(d) y_{t+1} - y_t = 3 \quad y_0 = 5$$

$$(a = -1) \quad (c = 3)$$

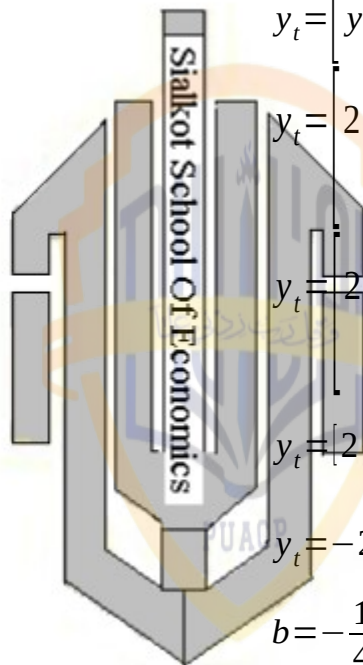
Applying formula

$$y_t = y_0 + ct \quad \text{as } a = -1$$

$$y_t = 5 + 3t$$

Since $b = -a = 1 > 0$ no oscillation will occur

$|b| = 1$ time path will be a horizontal straight line (i.e. divergent)



Assignment:

Q: Solve the following difference equation

(b) Check the answer, using $t=0$ and $t=1$

(c) Comment on the nature of the time path

1. $y_t = 6y_{t-1}$

2. $y_t = -\frac{1}{4}y_{t-1} + 60$ and $y_0 = 8$

3. $2y_t = y_{t-1} - 18$

4. $g_t = g_{t-1} - 25$ $g_0 = 40$

5. $5y_t + 2y_{t-1} - 140 = 0$; $y_0 = 30$

6. $x_t = 4x_{t-1} - 36$

7. $8y_{t-2} - 2y_{t-3} = 120$; $y_0 = 28$

Lagged Income Determination**Models:**

Find y_t and (b), (c) in above question

8. $c_t = 90 + 0.8y_{t-1}$; $I_t = 50$; $y_0 = 1200$

9. $c_t = 200 + 0.7y_{t-1}$; $I_t = 50 + 0.15y_{t-1}$; $y_0 = 3000$

10. $c_t = 0.92y_{t-1}$; $I_t = 0.08y_{t-1}$; $y_0 = 4000$

11. $c_t = 300 + 0.5y_t + 0.4y_{t-1}$; $I_t = 200 + 0.2y_{t-1}$; $y_0 = 6500$

12. $c_t = 200 + 0.5y_t$; $I_t = 3(y_t - y_{t-1})$; $y_0 = 10000$

The Cobweb Model:

The cobweb model differs from our earlier market models in that it treats Q_s as a function not of the current price but of the price of the preceding time period.

The Model:

Taking the linear versions of the (lagged) supply and (in lagged) demand functions and assuming that in each period the market price is always set at a level which clears the market, we have a market model with the following three equations:

$$Q_{dt} = Q_{st}$$

$$Q_{dt} = \alpha - \beta p_t \quad (\alpha, \beta > 0)$$

$$Q_{st} = -\alpha + \delta p_{t-1} \quad (\alpha, \delta > 0)$$

By substituting the last two equations into the first, however, the model can be reduced to a single first order difference equation as follows:

$$\alpha - \beta p_t = -\alpha + \delta + \delta p_{t-1}$$

$$-\beta p_t - \delta p_{t-1} = -\alpha - \gamma$$

$$\beta p_t + \delta p_{t-1} = \alpha + \gamma$$

Dividing through by β and shifting time subscripts ahead by one period [alter t to $(t+1)$, etc.]

$$p_{t+1} + \frac{\delta}{\beta} p_t = \frac{\alpha + \gamma}{\beta}$$

$$a = \frac{\delta}{\beta}; \quad c = \frac{\alpha + \gamma}{\beta}$$

$$p_t = \left[p_0 - \frac{c}{1+a} \right] (-a)^t + \frac{c}{1+a}$$

$$p_t = \left[p_0 - \frac{\alpha + \gamma}{\beta + \delta} \right] \left(-\frac{\delta}{\beta} \right)^t + \frac{\alpha + \gamma}{\beta + \delta}$$

Where p_0 represents the initial price

The Cobwebs:

Three points may be observed in regard to this time path. In the first place, the

expressions $\frac{(\alpha + \gamma)}{(\beta + \delta)}$, which constitute the particular integral of the difference equation, can be taken as the inter temporal equilibrium price of the model:

$$\bar{p} = \frac{\alpha + \gamma}{\beta + \delta}$$

Because this constant, it is a stationary equilibrium. Substituting $\bar{p}$ into our solution, we can express the time path alternatively in the form

$$p_t = (p_0 - \bar{p}) \left(-\frac{\delta}{\beta} \right)^t + \bar{p}$$

This leads us to the second point, namely the signification of the expression $(p_0 - \bar{p})$. Since this corresponds to the constant A in the Ab^t term, its sign will decide whether the time path will commence above or below the

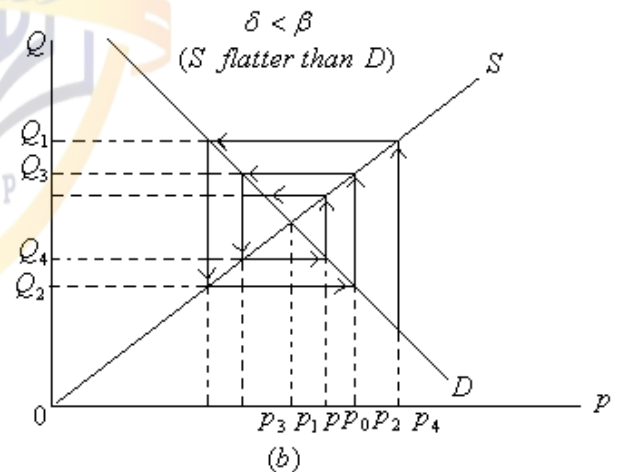
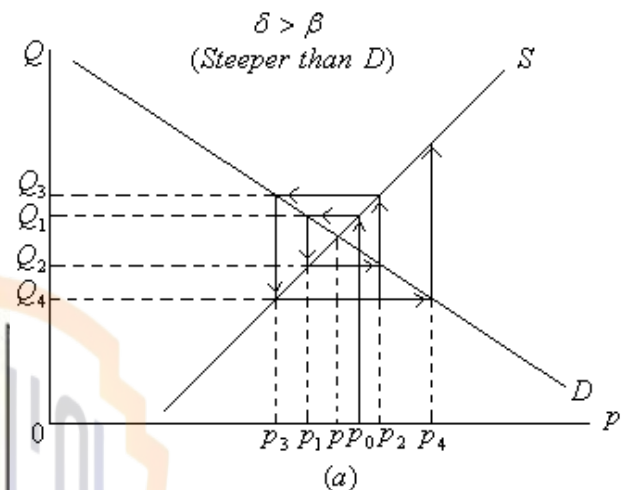
equilibrium (mirror effect), whereas its magnitude will decide how far above or below (scale effect). Lastly, there is the

expression $\left(-\frac{\delta}{\beta}\right)$ which corresponds to the b component of Ab^t . from our model specification that $(\beta, \delta > 0)$ we can deduce an oscillatory time path. It is this fact which gives rise to the cobweb phenomenon. There can of course arise there possible varieties of oscillatory of oscillation patterns in the model.

Explosive Uniform if $\delta > \beta$

The case of $\delta > \beta$ (S steeper than D) and the case of $\delta < \beta$ (S flatter than D) are illustrated in diagram a and b respectively. In either case, however the interaction of D and S will yield the intertemporal equilibrium price $\bar{P}$. When $\delta > \beta$, as in diagram a, the interaction of demand and supply will produce an explosive oscillation as follows. Given an initial price p_0 (here assumed above $\bar{P}$) we can follow the arrow head and read off on the δ curve that the quantity supplied in the next period (period 1) will be Q_1 . In order to clear the market, the quantity demand in period 1 must also be Q_1 which is possible if and only if price p_1 will lead to Q_2 as the quantity supplied in period 2 and to clear the market in the latter period, price must be set at the level of p_2 according to demand curve. This process will continue and will create a cobweb which is centrifugal with the cobweb being spun from inside out, the time path is divergent and the oscillation explosive.

By way of contrast, in the case of diagram b, where $\delta < \beta$, the spinning process will create a cobweb which is centripetal. From p_0 if we follow the arrow heads we shall be led over closed to the interaction of the demand and supply curve, where $\bar{P}$ is while still oscillatory this price path is convergent.



Exercise 16.4:

Q.1. On the basis of (1.1.), find the time path of Q and analyze the conditions for its convergence.

$$Q_{dt} = Q_{st} \quad (i)$$

$$Q_{dt} = \alpha - \beta p_t \quad (ii) \quad (16.10)$$

$$Q_{st} = -\gamma + \delta p_{t-1} \quad (iii)$$

Substituting the time subscripts ahead by one period and rearranging the equation

$$p_{t+1} + \frac{\delta}{\beta} p_t = \frac{\alpha + \gamma}{\beta} \quad a = \frac{\delta}{\beta}$$

$$c = \frac{\alpha + \gamma}{\beta}$$

Applying formula As $a \neq -1$

$$p_t = \left[p_0 - \frac{c}{1+a} \right] (-a)^t + \frac{c}{1+a}$$

$$p_t = \left[p_0 - \frac{\alpha + \gamma / \beta}{1 + \delta / \beta} \right] \left(-\frac{\delta}{\beta} \right)^t + \frac{\alpha + \gamma}{1 + \frac{\delta}{\beta}}$$

$$p_t = \left[p_0 - \frac{\alpha + \gamma}{\beta + \delta} \right] \left(-\frac{\delta}{\beta} \right)^t + \frac{\alpha + \gamma}{\beta + \delta}$$

Putting p_t in equation (ii)

$$Q_{dt} = Q_{st} = Q$$

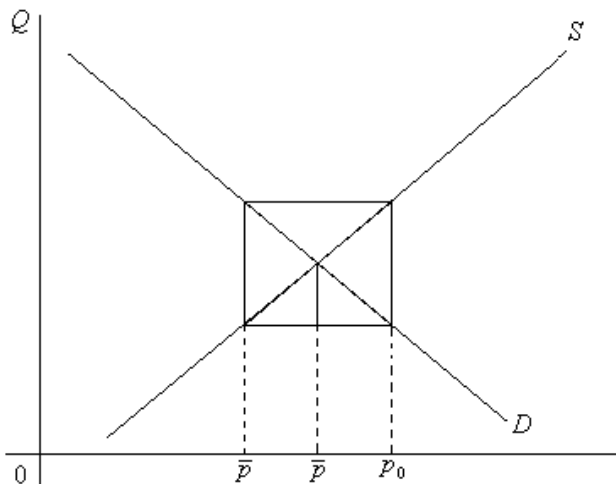
$$Q = \alpha - \beta p_t$$

$$Q = \alpha - \beta \left[p_0 - \frac{\alpha + \gamma}{\beta + \delta} \right] \left(-\frac{\delta}{\beta} \right)^t - \beta \left[\frac{\alpha + \gamma}{\beta + \delta} \right]$$

As $\delta, \beta > 0$ the time path will be convergent if and only if

$$|\delta| < |\beta| \text{ or } \delta < \beta$$

Q.2. Draw a diagram similar to those of figure 16.2 to show that for the case of $\delta = \beta$ the price will oscillate uniformly with neither damping nor explosion.



Q.3. Given the demand and supply for cobweb model as follows, find the inter temporal equilibrium price and determine whether the equilibrium is stable

$$(a) Q_{dt} = 18 - 3 p_t$$

$$Q_{st} = -3 + 4 p_{t-1}$$

$$Q_{dt} = Q_{st}$$

$$18 - 3 p_t = -3 + 4 p_{t-1}$$

$$3 p_t + 4 p_{t-1} = 18 + 3$$

$$3 p_t + 4 p_{t-1} = 21$$

Shifting the time subscripts ahead by one period and dividing through by 3

$$p_{t+1} + \frac{4}{3} p_t = 7$$

$$a = \frac{4}{3}; \quad c = 7$$

Since $a \neq -1$ applying formula

$$p_t = \left[p_0 - \frac{c}{1+a} \right] (-a)^t + \frac{c}{1+a}$$

$$p_t = \left[p_0 - \frac{7}{1+4/3} \right] (-4/3)^t + \frac{7}{1+4/3}$$

$$p_t = [p_0 - 3] (-4/3)^t + 3$$

Hence inter temporal equilibrium price is

$$\bar{p} = 3$$

Since $b = -\frac{4}{3} < 0$ oscillation will occur

$|b| = \frac{4}{3} > 1$ Time path will be divergent.

Hence unstable equilibrium

$$(b) Q_{dt} = 22 - 3 p_t$$

$$Q_{st} = -2 + p_{t-1}$$

At equilibrium

$$Q_{dt} = Q_{st}$$

$$22 - 3 p_t = -2 + p_{t-1}$$

$$3 p_t + p_{t-1} = 22 + 2$$

$$3 p_t + p_{t-1} = 24$$

Shifting time subscripts ahead by one period and dividing through by 3

$$p_{t+1} + \frac{1}{3} p_{t+1} = 8 \quad \text{Since } a = \frac{1}{3}$$

$$c=8$$

As $a \neq -1$ Applying formula

$$p_t = \left[p_0 - \frac{c}{1+a} \right] (-a)^t + \frac{c}{1+a}$$

$$p_t = \left[p_0 - \frac{8}{1+\frac{1}{3}} \right] \left(-\frac{1}{3} \right)^t + \frac{8}{1+\frac{1}{3}}$$

$$p_t = \left[p_0 - \frac{8}{\frac{4}{3}} \right] \left(-\frac{1}{3} \right)^t + \frac{8}{\frac{4}{3}}$$

$$p_t = [p_0 - 6] \left(-\frac{1}{3} \right)^t + 6$$

Hence the inter temporal equilibrium is $\bar{p}=6$

Since $b = -\frac{1}{3} < 0$

$$|b| = \frac{1}{3} < 1$$

Time path shows damped oscillation equilibrium will be stable

(c) $Q_{dt} = 19 - 6p_t$

$$Q_{st} = 6p_{t-1} - 5$$

At equilibrium

$$Q_{dt} = -Q_{st}$$

$$Q_{dt} = Q_{st}$$

$$19 - 6p_t = 6p_{t-1} - 5$$

$$6p_t + 6p_{t-1} = 19 + 5$$

$$6p_t + 6p_{t-1} = 24$$

Shifting time subscripts ahead by one period and dividing through by 6

$$p_t + p_{t-1} = 4 \quad a=1$$

$$c=4$$

As $a \neq -1$ applying formula

$$p_t = \left[p_0 - \frac{c}{1+a} \right] (-a)^t + \frac{c}{1+a}$$

$$p_t = \left[p_0 - \frac{4}{1+1} \right] (-1)^t + \frac{4}{1+1}$$

$$p_t = [p_0 - 2] (-1)^t + 2$$

Hence the inter temporal equilibrium price is

$$\bar{p}=2$$

Since $b = -1 < 0$

$$|b|=1$$

The time path will show uniform oscillation and equilibrium will be unstable

Q.4. Given the model

$$Q_{dt} = Q_{st}$$

$$Q_{dt} = \alpha - \beta p_t$$

$$Q_{st} = -\gamma + \delta p_t^e$$

Where p_t^e denotes the expected price for period t. furthermore, suppose that sellers have the “adaptive” type of price expectation

$$p_t^e = p_{t-1}^e + \eta(p_{t-1} - p_{t-1}^e) \quad (0 < \eta \leq 1)$$

Where η is an expectation-adjustment coefficient

(a) Given an economic interpretation to the above equation. In what respect it is similar to and different from the adaptive expectations equation (15.34)?

Economic Interpretation:

$$p_t^e = p_{t-1}^e + \eta(p_{t-1} - p_{t-1}^e) \quad (0 < \eta \leq 1)$$

We can interpret this equation in three ways

1: when actual price for period t-1 becomes equal to expected price for period t-1, then expected price for period t will be equal to the expected price for period t-1 i.e.

When

$$p_{t-1} = p_{t-1}^e \quad \text{then} \quad p_t^e = p_{t-1}^e$$

2. When actual price for period t-1 exceed the expected price for period t-1, the expected price for period t will be greater than expected price for period t-1. In other words the expected price for period t will increase

When



$p_{t-1} > p_t^e$ then $p_t^e > p_{t-1}^e$

3: when expected price for period t-1 exceeds the actual price for the period t-1, the expected price for the period t will be smaller than the expected price for the period t-1. In other words we can say the expected price for period t will decrease if

$p_{t-1} < p_t^e$ then $p_t^e < p_{t-1}^e$

(b) What happens if η (eta) takes its maximum value? Can we consider the cobweb model as a special case of the present model?

$$Q_{dt} = \alpha - \beta p_t \quad (i) \quad (\alpha, \beta > 0)$$

$$Q_{st} = -\gamma + \delta p_t^e \quad (ii) \quad (\gamma, \delta > 0)$$

$$p_t^e = p_{t-1}^e + \eta(p_{t-1} - p_{t-1}^e) \quad (0 < \eta \leq 1)$$

The maximum value of η is 1. If η takes its max value, i.e. $\eta = 1$ then we can write

$$p_t^e = p_{t-1}^e + 1(p_{t-1} - p_{t-1}^e)$$

$$p_t^e = p_{t-1}^e + p_{t-1} - p_{t-1}^e$$

$$p_t^e = p_{t-1}$$

Putting $p_t^e = p_{t-1}$ in equation (ii), supply function becomes

$$Q_{st} = -\gamma + \delta p_{t-1}$$

Now we have

$$Q_{dt} = \alpha - \beta p_t$$

$$Q_{st} = -\gamma + \delta p_{t-1}$$

Therefore we can consider the cobweb model as a special case of the present model.

(c) Show that the new model can be represented by the first-order difference equation.

$$p_{t-1} - \left(1 - \eta - \frac{\eta\delta}{\beta}\right) p_t = \frac{\eta(\alpha + \gamma)}{\beta}$$

Solving the supply function for p_t^e

$$Q_{st} = -\gamma + \delta p_t^e$$

$$\delta p_t^e = Q_{st} + \gamma$$

$$p_t^e = \frac{Q_{st}}{\delta} + \frac{\gamma}{\delta}$$

Substituting this information that

$$Q_{st} = Q_{dt} = \alpha - \beta p_t$$

$$p_t^e = \frac{\alpha - \beta p_t}{\delta} + \frac{\gamma}{\delta}$$

Similarly

$$p_{t-1}^e = \frac{\alpha - \beta p_{t-1}}{\delta} + \frac{\gamma}{\delta}$$

Putting values of p_t^e and p_{t-1}^e in the equation given below

$$p_t^e = p_{t-1}^e + \eta(p_{t-1} - p_{t-1}^e) \quad 0 < \eta \leq 1$$

$$\frac{\alpha - \beta p_t}{\delta} + \frac{\gamma}{\delta} = \frac{\alpha - \beta p_{t-1}}{\delta} + \frac{\gamma}{\delta} + \eta \left(p_{t-1} - \left(\frac{\alpha - \beta p_{t-1}}{\delta} + \frac{\gamma}{\delta} \right) \right)$$

$$\frac{\alpha}{\delta} - \frac{\beta}{\delta} p_t + \frac{\gamma}{\delta} = \frac{\alpha}{\delta} - \frac{\beta}{\delta} p_{t-1} + \frac{\gamma}{\delta} + \eta p_{t-1} - \eta \frac{\alpha}{\delta} + \eta \frac{\beta p_{t-1}}{\delta} - \frac{\eta \gamma}{\delta}$$

$$-\frac{\beta}{\delta} p_t = -\frac{\beta}{\delta} p_{t-1} + \eta p_{t-1} + \eta \frac{\beta p_{t-1}}{\delta} - \frac{\eta}{\delta} (\alpha + \gamma)$$

Multiplying each term by δ

$$-\beta p_t = -\beta p_{t-1} + \eta \delta p_{t-1} + \eta \beta p_{t-1} - \eta (\alpha + \gamma)$$

Dividing each term by $-\beta$

$$p_t = p_{t-1} - \eta \frac{\delta}{\beta} p_{t-1} - \eta p_{t-1} + \eta \frac{(\alpha + \gamma)}{\beta}$$

$$p_t = p_{t-1} - \eta p_{t-1} - \frac{\eta \delta}{\beta} p_{t-1} + \eta \frac{(\alpha + \gamma)}{\beta}$$

$$p_t = \left(1 - \eta - \frac{\eta \delta}{\beta} \right) p_{t-1} + \eta \frac{(\alpha + \gamma)}{\beta}$$

$$p_t - \left(1 - \eta - \frac{\eta \delta}{\beta} \right) p_{t-1} = \eta \frac{(\alpha + \gamma)}{\beta}$$

Shifting the time subscripts ahead by one period

$$p_{t+1} - \left(1 - \eta - \frac{\eta \delta}{\beta} \right) p_t = \eta \frac{(\alpha + \gamma)}{\beta}$$

(d) Find the time path of price. Is path necessarily oscillatory? Can it be oscillatory? Under what circumstances

$$Q_{dt} = \alpha - \beta p_t \quad (\alpha, \beta > 0)$$

$$Q_{st} = -\gamma + \delta p_t^e \quad (\gamma, \delta > 0)$$

$$Q_{dt} = Q_{st}$$

$$\alpha - \beta p_t = -\gamma + \delta p_t^e \quad (i)$$

$$p_{t+1} - \left(1 - \eta - \eta \frac{\delta}{\beta}\right) p_t = \eta \frac{(\alpha + \gamma)}{\beta} \quad (ii)$$

This is the first order linear differences equation.

Where

$$c = \eta \frac{(\alpha + \gamma)}{\beta} ; \quad a = -\left(1 - \eta - \eta \frac{\delta}{\beta}\right)$$

$$p_t = \left[p_0 - \frac{c}{1+a} \right] (-a)^t + \frac{c}{1+a}$$

$$p_t = \left[p_0 - \frac{\frac{\eta(\alpha + \gamma)}{\beta}}{1 - \left(1 - \eta - \eta \frac{\delta}{\beta}\right)} \right] \left(1 - \eta - \eta \frac{\delta}{\beta}\right)^t + \frac{\frac{\eta(\alpha + \gamma)}{\beta}}{1 - \left(1 - \eta - \eta \frac{\delta}{\beta}\right)}$$

$$p_t = \left[p_0 - \frac{\frac{\eta(\alpha + \gamma)}{\beta}}{1 - 1 + \eta + \eta \frac{\delta}{\beta}} \right] \left(1 - \eta - \eta \frac{\delta}{\beta}\right)^t + \frac{\frac{\eta(\alpha + \gamma)}{\beta}}{1 - 1 + \eta + \eta \frac{\delta}{\beta}}$$

$$p_t = \left[p_0 - \frac{\frac{\eta(\alpha + \gamma)}{\beta}}{\eta\beta + \frac{\eta\delta}{\beta}} \right] \left(1 - \eta - \eta \frac{\delta}{\beta}\right)^t + \frac{\frac{\eta(\alpha + \gamma)}{\beta}}{\eta\beta + \frac{\eta\delta}{\beta}}$$

$$p_t = \left[p_0 - \frac{\eta(\alpha + \gamma)}{\eta(\beta + \delta)} \right] \left(1 - \eta - \eta \frac{\delta}{\beta}\right)^t + \frac{\eta(\alpha + \gamma)}{\eta(\beta + \delta)}$$

$$p_t = \left[p_0 - \frac{(\alpha + \gamma)}{(\beta + \delta)} \right] \left(1 - \eta - \eta \frac{\delta}{\beta}\right)^t + \frac{(\alpha + \gamma)}{(\beta + \delta)}$$

At equilibrium $p_t = p_t^e = \bar{p}$ putting $\bar{p}$ in

place of p_t and p_t^e in (i)

$$\alpha - \beta \bar{p} = -\gamma + \delta \bar{p}$$

$$\beta \bar{p} + \delta \bar{p} = \alpha + \gamma$$

$$\bar{p}(\beta + \delta) = \alpha + \gamma$$

$$\bar{p} = \frac{\alpha + \gamma}{\beta + \delta}$$

$$\therefore p_t = [p_0 - \bar{p}] \left(1 - \eta - \eta \frac{\delta}{\beta}\right)^t + \bar{p}$$

This time path is not necessarily oscillatory. It can be oscillatory if $\eta + 1$ As $\delta, \beta > 0$

(e) Show that the time path p_t if oscillatory will converge only if

$$\frac{1-2}{\eta} < -\frac{\delta}{\beta}$$

As compared with the cobweb solution does the new model have a wider or narrower range for the stability inducing values of $-\frac{\delta}{\beta}$?

$$p_t = \left[p_0 - \frac{\alpha + \gamma}{\beta + \delta} \right] \left(1 - \eta - \eta \frac{\delta}{\beta}\right)^t + \frac{\alpha + \gamma}{\beta + \delta}$$

p_t will converge only if

$$\left|1 - \eta - \eta \frac{\delta}{\beta}\right| < 1$$

$$\left| -1 + \eta + \eta \frac{\delta}{\beta} \right| < 1$$

$$-1 + \eta + \eta \frac{\delta}{\beta} < 1$$

$$\eta + \eta \frac{\delta}{\beta} < 1 + 1$$

$$\eta \left(1 + \frac{\delta}{\beta}\right) < 2$$

$$1 + \frac{\delta}{\beta} < \frac{2}{\eta}$$

Or

$$\frac{2}{\eta} > 1 + \frac{\delta}{\beta}$$

$$\frac{2}{\eta} - 1 > \frac{\delta}{\beta}$$

$$-\left(\frac{2}{\eta}-1\right) < -\frac{\delta}{\beta}$$

$$1 - \frac{2}{\eta} < -\frac{\delta}{\beta}$$

Thus it is proved that p_t will converge

$$\text{only if } \frac{1-2}{\eta} < -\frac{\delta}{\beta}$$

This model has a narrower range for the

stability inducing values of $-\frac{\delta}{\beta}$

Q.5. the cobweb model like the previously encountered dynamic market models is essentially based on the static market model presented in section 3.2.

What economic assumption is the dynamizing agent in the present case? Explain.

Answer:

The assumption that the producers output decisions must be made one period in advance of the actual sale i.e. supply is a function of lagged period price is the dynamizing agent in the present case.

